

Network Security

Lecture 5: Encryption





Learning objectives

- Define the cryptography concepts,
- Understand Caesar cipher substitution technique,
- Explain symmetric and asymmetric encryption algorithms,
- Explain Diffie-Hellman Key Exchange,
- Explain Hash functions,
- Explain keyed-hash message authentication code,
- Explain the basics of cryptanalysis.



Cryptography Basics



Cryptography is the science of secret or hidden writing



It has two main components:

Encryption

- Practice hiding messages so that they can not be read by anyone other than the intended recipient

Authentication & Integrity

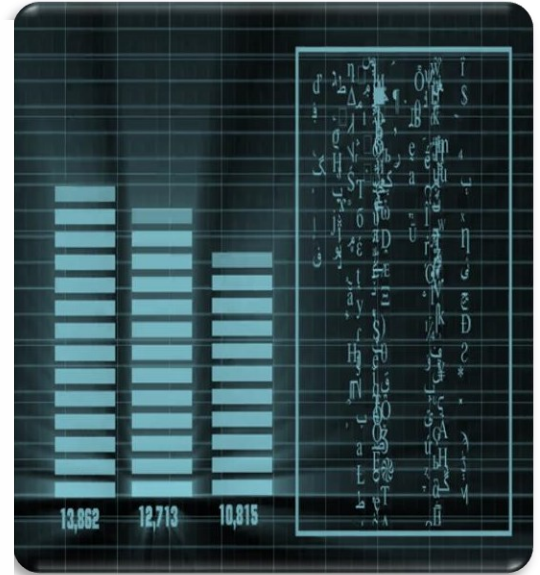
- Ensuring that users of data/resources are the persons they claim to be and that a message has not been altered

Types of Encryption Algorithms



Encryption and
decryption
use the same
key

Uses different
keys for
encryption
and decryption.



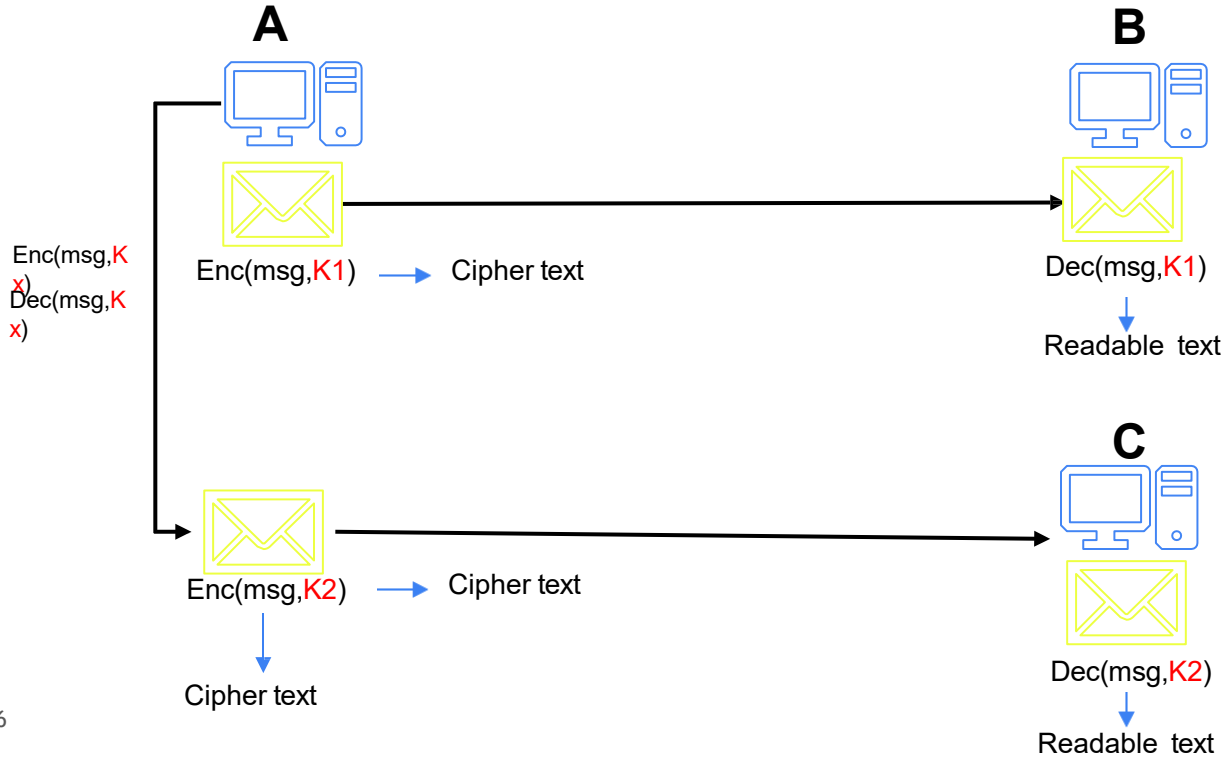


AES

DES

IDEA





Asymmetric Encryption Algorithms

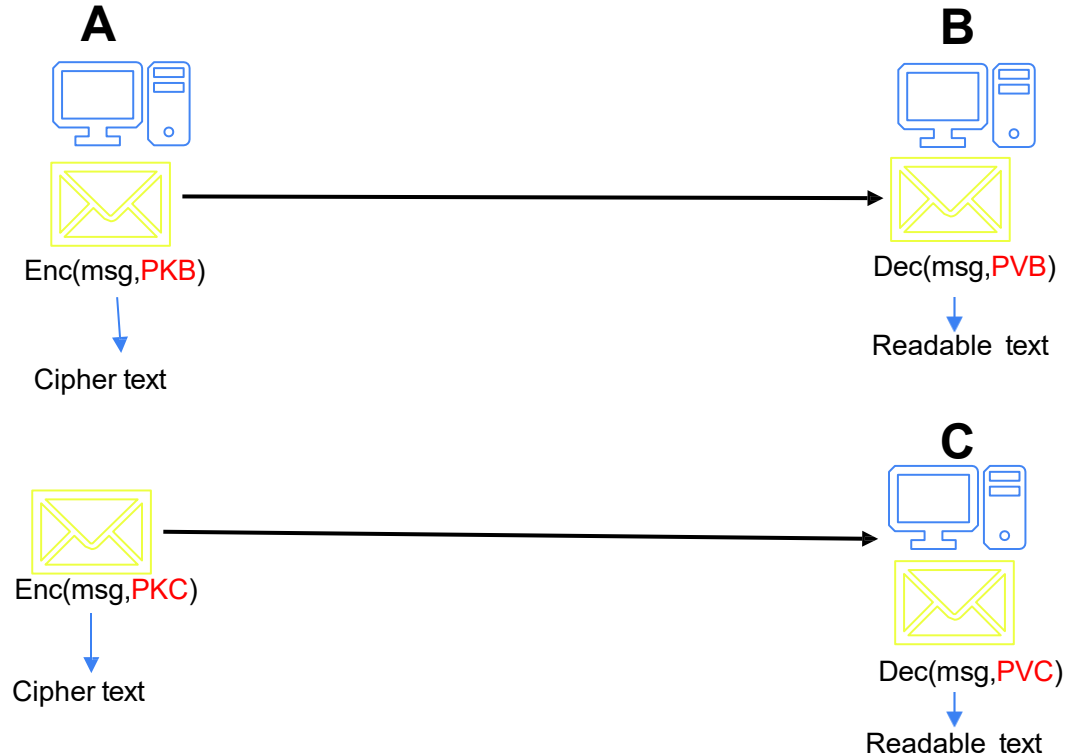


Elliptic curve

Digital
signature



Asymmetric Key Cryptography



PVC = private key C
PVB = private key B
PKC = public key C
PKB = public key B



Encryption - Symmetric Algorithms



Algorithms in which the key for encryption and decryption are the same are Symmetric

Example: Caesar Cipher



Types:

Block Ciphers

- Encrypt data one block at a time (typically 64 bits, or 128 bits)
- Used for a single message

Stream Ciphers

- Encrypt data one bit or one byte at a time

Symmetric Encryption - Key Strength



Strength of the algorithm is determined by the size of the key

The longer the key the more difficult it is to crack



Key length is expressed in bits

Typical key sizes vary between 48 bits and 448 bits



Set of possible keys for a cipher is called keys-space

For a 40-bit key there are 2^{40} possible keys
For 128-bit key there are 2^{128} possible keys
Each additional bit added to the key length doubles the security

Symmetric Encryption - Key Strength (Cont.)

- To crack the key the hacker must use brute-force

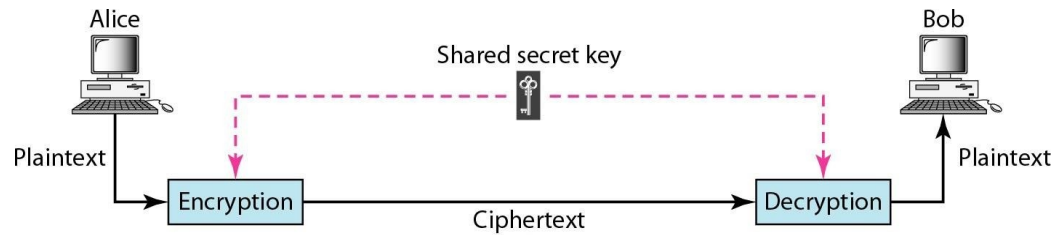
(i.e. try all the possible keys till a key that works is found)

- Supercomputer can crack a 56-bit key in 24 hours
- It will take 2^{72} times longer to crack a 128-bit key (Longer than the age of the universe)

* A supercomputer is a computer with a high level of performance as compared to a general-purpose computer



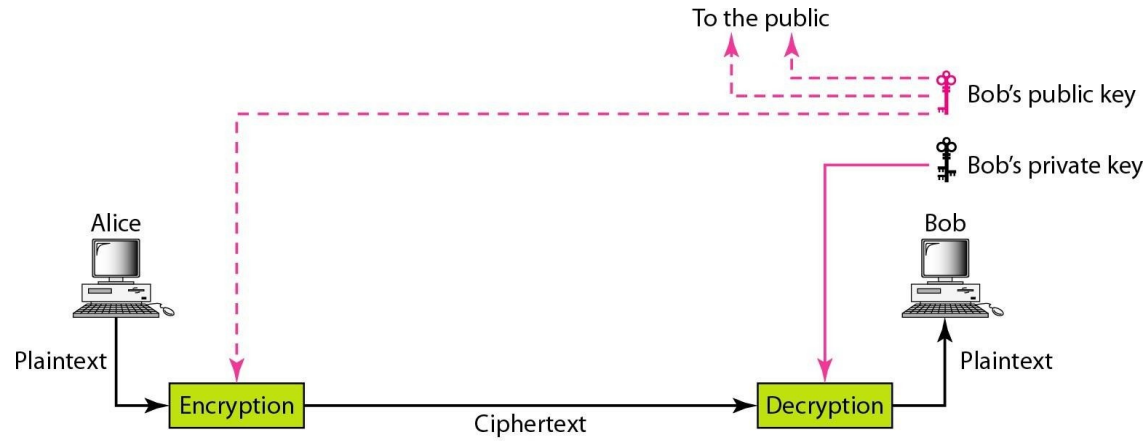
Symmetric-key cryptography



Symmetric-key cryptography

- Advantages:
 - Simple
 - Faster
- Disadvantages:
 - Key must be exchanged in secure way
 - Easy for hacker to get a key as it is passed in unsecure way.

Figure 30.4 *Asymmetric-key cryptography*





Note

An asymmetric-key (or public-key) cipher uses two keys: one public (To encrypt data) and one private (To decrypt data).

- **Asymmetric Key Cryptography (Public Key Cryptography)**
 - 2 different keys are used
 - Users get the Key from an Certificate Authority

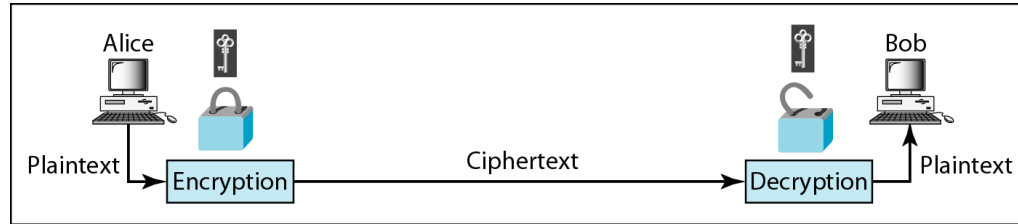
Advantages

1. **More Secured**
2. **Authentication**

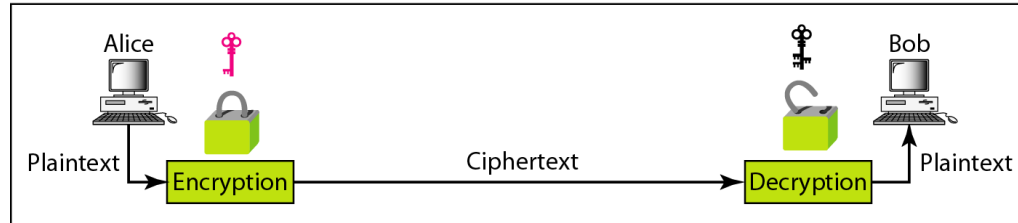
Disadvantages

1. **Relatively Complex**

Figure 30.6 *Comparison between two categories of cryptography*



a. Symmetric-key cryptography



b. Asymmetric-key cryptography

Symmetric key crypto

- DES
- AES
- CBC

Symmetric key crypto: DES

DES: Data Encryption Standard

- US encryption standard [NIST 1993]
- 56-bit symmetric key, 64 bit plaintext input
- How secure is DES?
 - DES Challenge: 56-bit-key-encrypted phrase (“Strong cryptography makes the world a safer place”) decrypted (brute force) in 4 months
 - no known “backdoor” decryption approach
- making DES more secure
 - use three keys sequentially (3-DES) on each datum
 - use cipher-block chaining

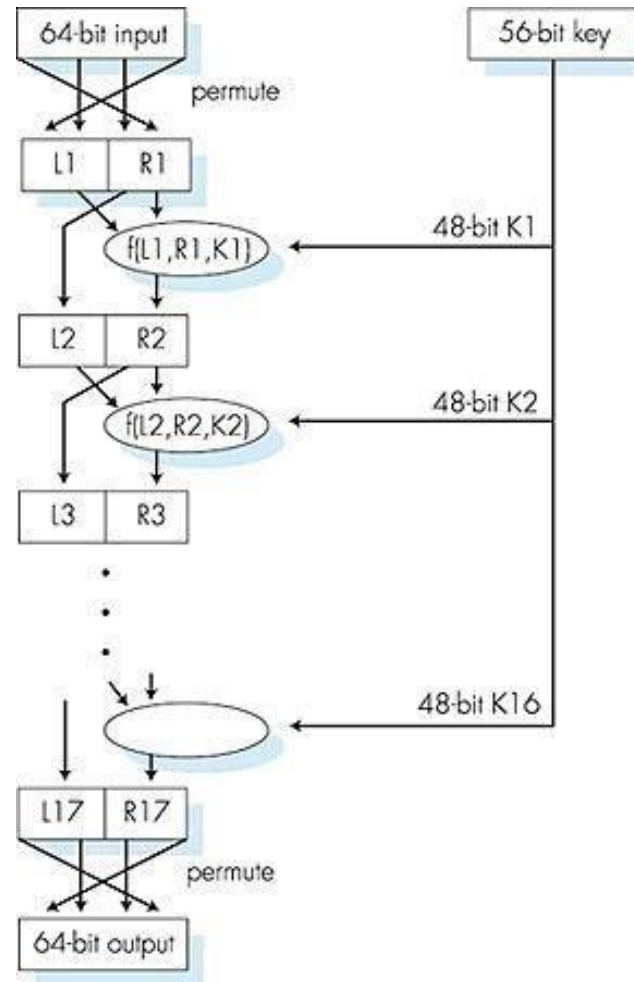
Symmetric key crypto: DES

DES operation

initial permutation

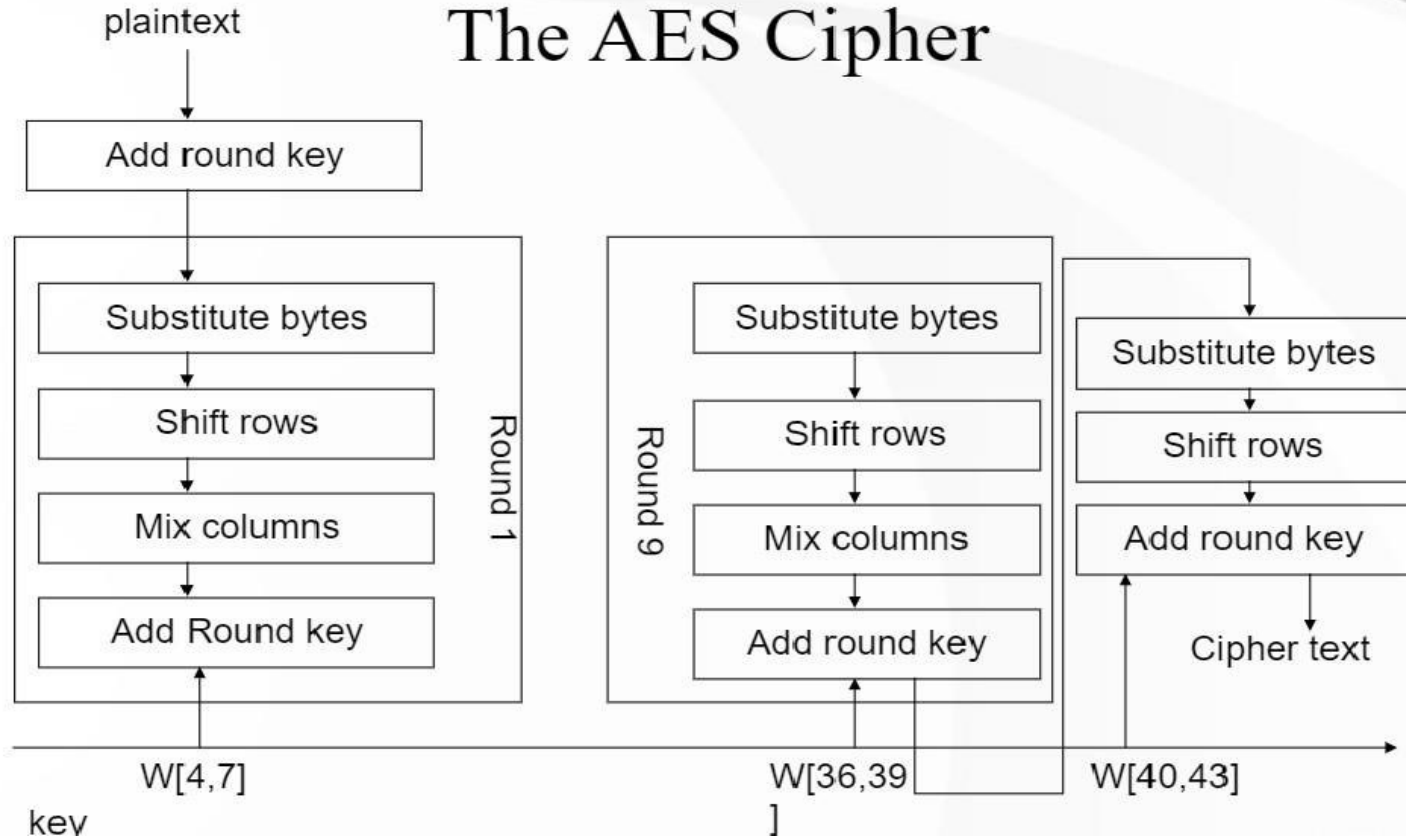
16 identical “rounds” of
function application,
each using different 48
bits of key

final permutation

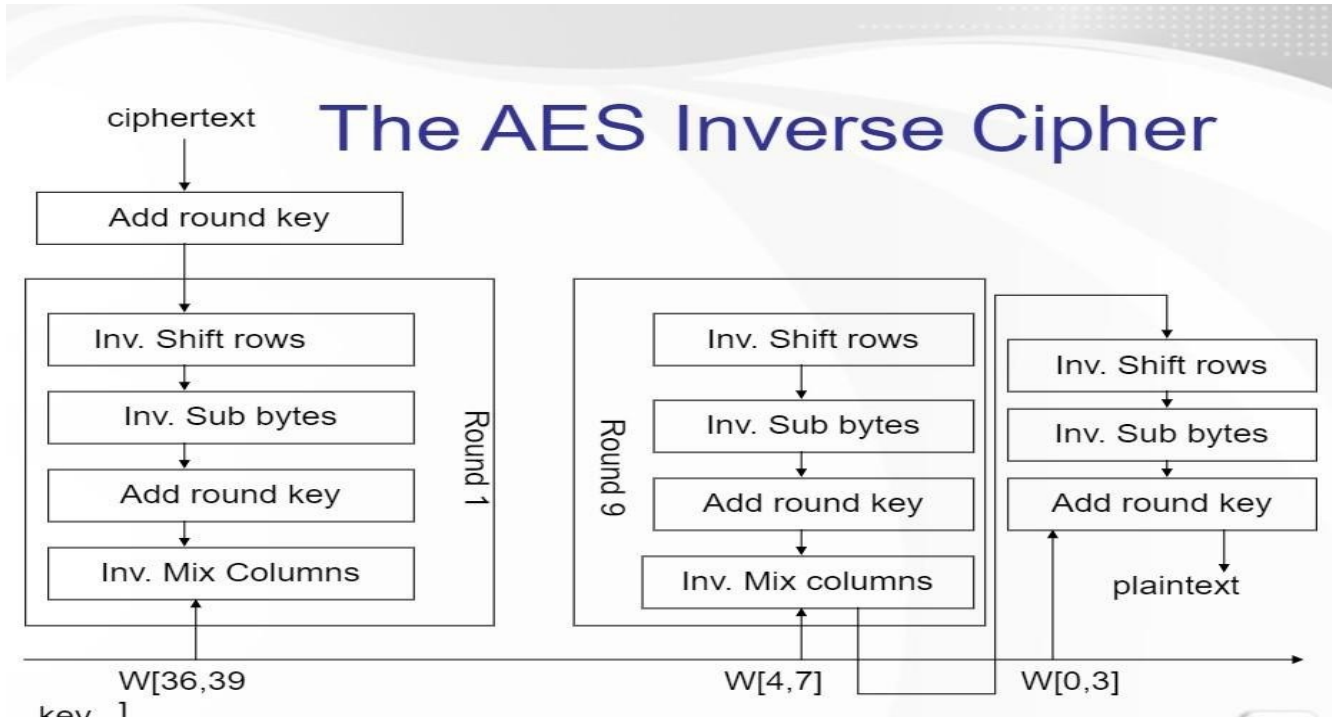


AES Algorithm

The AES Cipher

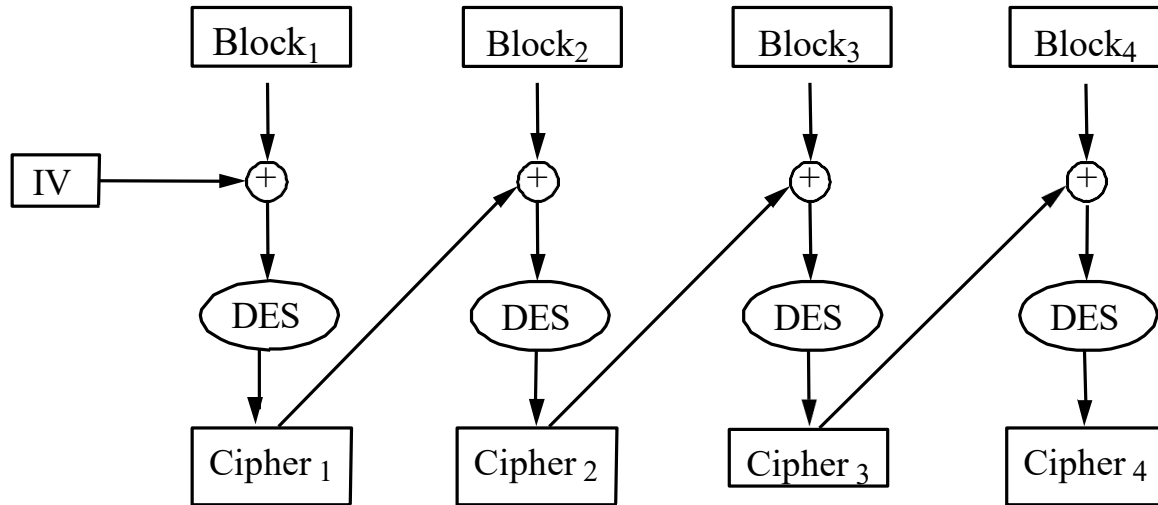


AES DECRYPTION



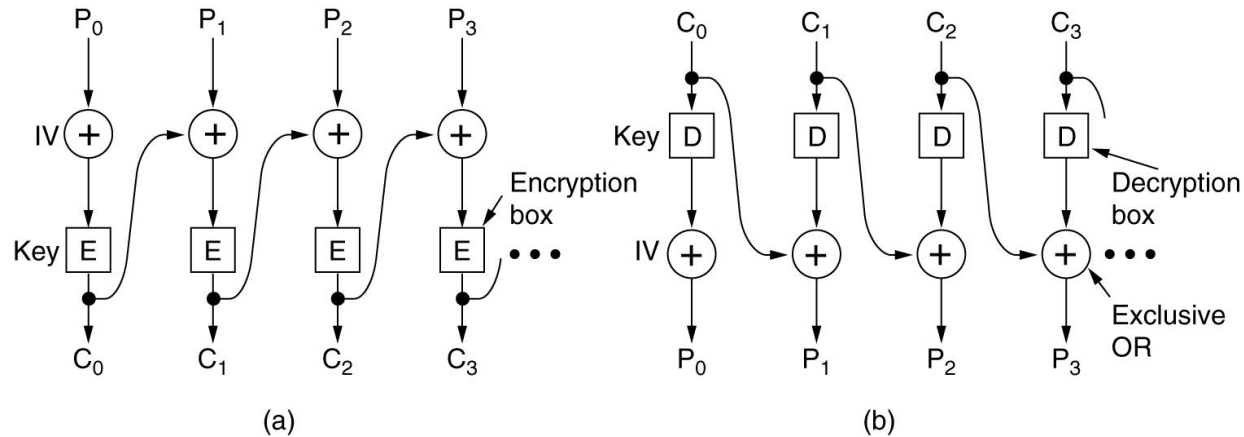
Cipher block chaining (CBC)

- Break plaintext into blocks
- XOR previous round ciphertext with new plaintext



Cipher Block Chaining Mode

- Cipher block chaining. (a) Encryption. (b) Decryption.





Substitution Ciphers - Caesar Cipher

- Caesar Cipher is a method in which each letter in the alphabet is rotated by **three** letters.
- Caesar cipher substitution technique is vulnerable to **brute-force** cryptanalysis because its key space is small, i.e., there are only 25 possible keys. In addition, the Caesar cipher substitution technique is vulnerable to **frequency analysis**.

A B C D E F G H I J K L M N O P Q R S T U V W X Y Z



D E F G H I J K L M N O P Q R S T U V W X Y Z A B C





Substitution Ciphers - Caesar Cipher (cont.)

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25
A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C

•Let us try to encrypt the message

ATTACK AT DAWN

DWWDFN DW GDZQ



[Frequency analysis](#)





Substitution Cipher - Monoalphabetic Cipher

- Monoalphabetic cipher is a substitution cipher in which for a given key, the cipher alphabet for each plain alphabet is fixed throughout the encryption process. For example, **if 'A' is encrypted as 'D', for any number of occurrences in that plaintext, 'A' will always get encrypted to 'D'.**
- Any letter can be substituted for any other letter. Each letter must have a unique substitute
- The **monoalphabetic substitution** consists in using a **mixed alphabet** (with the letters in an unusual order) and **replacing the letters** of the alphabet normal with it.
- The key is changed for every message (possibly 26!).

ABCDEFGHIJKLMNOPQRSTUVWXYZ
MNBVCXZASDFGHJKLPOIUYTREWQ



Substitution Cipher - Polyalphabetic Caesar Cipher

Developed by Blaise de Vigenere. Also, called Vigenere cipher

Vigenere Cipher is a method of encrypting alphabetic text.

It uses a simple form of polyalphabetic substitution. A polyalphabetic cipher is any cipher based on substitution, using multiple substitution alphabets.

The encryption of the original text is done using the Vigenère square or Vigenère table.

Substitution Cipher - Polyalphabetic Caesar Cipher (cont.)

Encryption

Plaintext: ATTACKATDAWN

Key: Lemon

Keystream: lemonlemonle

Cipher text:

LXFOPVEFRNHR

Key

Plain text

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
A	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
B	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A
C	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B
D	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C
E	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D
F	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E
G	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F
H	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G
I	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H
J	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I
K	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J
L	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K
M	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L
N	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M
O	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N
P	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
Q	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P
R	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q
S	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R
T	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S
U	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T
V	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U
W	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V
X	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W
Y	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X
Z	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y

Vigenère square or Vigenère table

Substitution Cipher - Polyalphabetic Caesar Cipher (cont.)

Decryption

Key: Lemon

Cipher text:

LXFOPVEFRNHR

Keystream: lemonlemonle

Plaintext: ATTACKATDAWN

Key

Plain text

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
A	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
B	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A
C	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B
D	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C
E	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D
F	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E
G	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F
H	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G
I	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H
J	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I
K	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J
L	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K
M	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L
N	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M
O	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N
P	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
Q	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P
R	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q
S	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R
T	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S
U	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T
V	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U
W	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V
X	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W
Y	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X
Z	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y

Vigenère square or Vigenère table

Symmetric Encryption - Limitations

Any exposure to the secret key compromises the secrecy of the ciphertext.

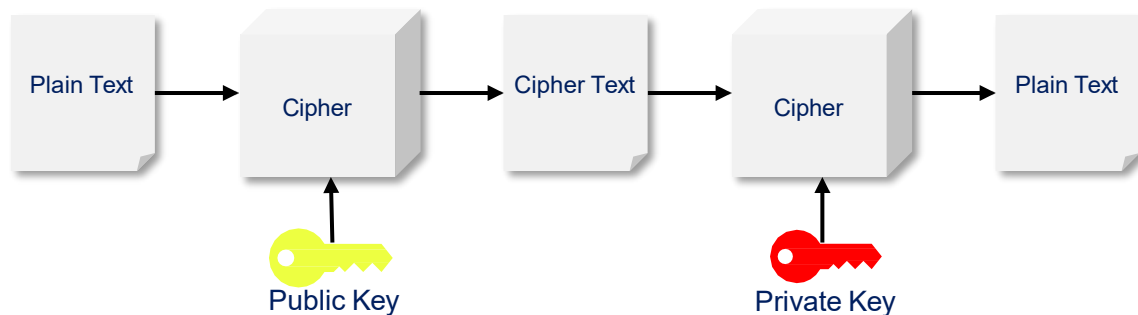
A key needs to be delivered to the recipient of the coded message for it to be decrypted.

Potential for **eavesdropping** attack during transmission of key.



Encryption - Asymmetric Algorithms

- Uses a pair of keys for encryption
 - Public key for encryption
 - Private key for decryption
- Messages encoded using the public key can only be decoded by the private key
 - Secret transmission of key for decryption is not required
 - Every entity can generate a key pair and release its public key



Asymmetric Encryption - Types

- Two most popular algorithms are RSA & El Gamal
- RSA
- Developed by Ron Rivest, Adi Shamir, Len Adelman
- Both public and private key are interchangeable
- Variable Key Size (512, 1024, or 2048 bits)
- Most popular public key algorithm
- El Gamal
- Developed by Taher ElGamal
- Variable key size (512 or 1024 bits)
- Less common than RSA, used in protocols like PGP



Asymmetric Encryption - Weaknesses



Efficiency is lower than Symmetric Algorithms. It's a slow process compared to symmetric cryptography. Therefore, it's not appropriate for decrypting bulk messages.

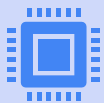


Potential for a man-in-the-middle attack



Since public keys aren't authenticated, nobody can be certain that a public key belongs to the person supplied.

Diffie-Hellman



Diffie-Hellman is the first key agreement algorithm

Invented by Whitfield Diffie & Martin Hellman

Provided the ability for messages to be exchanged securely without having to have shared some secret information previously

Inception of public key cryptography which allowed keys to be exchanged in the open



No exchange of secret keys

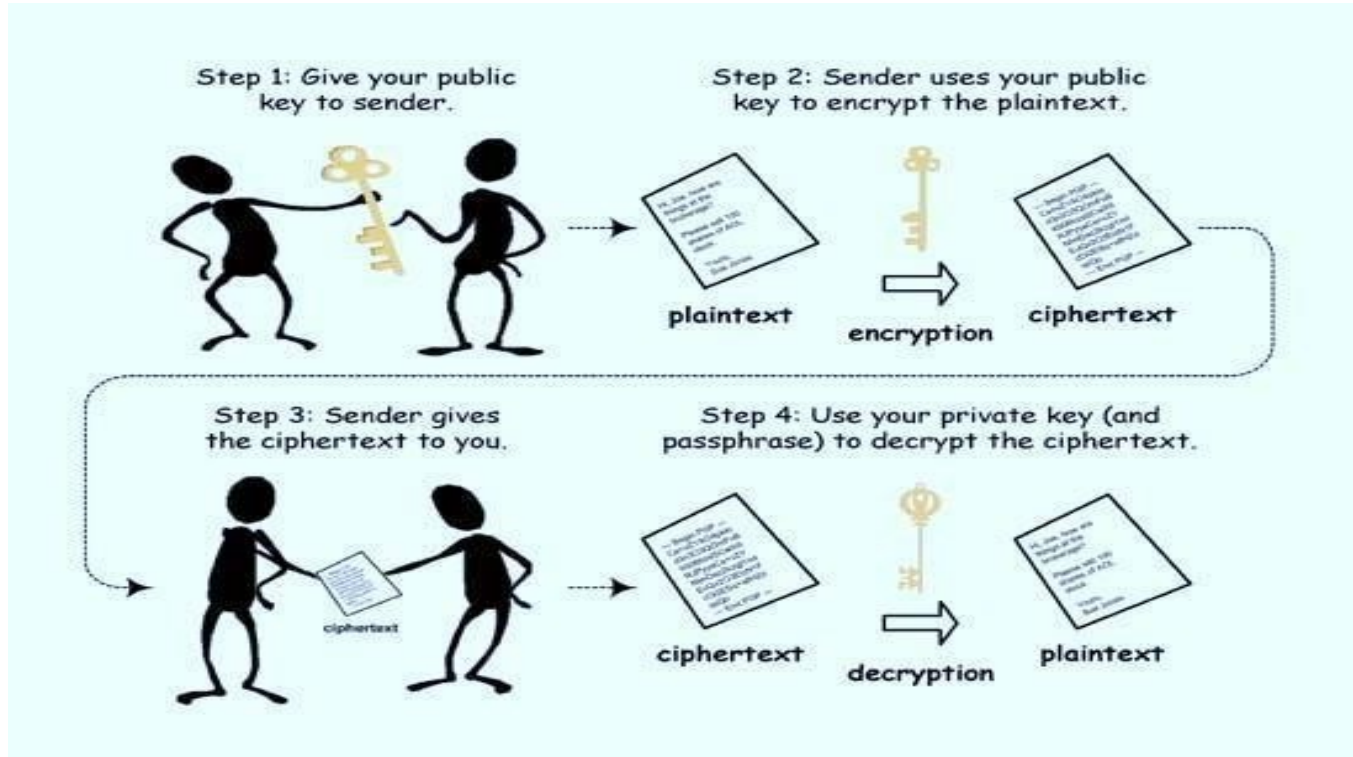
Man-in-the-middle attack avoided



ASYMMETRIC ENCRYPTION

- Asymmetric encryption use two keys, one to encrypt the data, and another key to decrypt the data.
- These keys are generated together
- . One is named as Public key and is distributed freely. The other is named as Private Key and it is kept hidden.
- Both Sender & Recipient has to share their Public Keys for Encryption and has to use their Private Keys for Decryption.

How it WORKS.....?



Key Points in Asymmetric Encryption

- ❖ Asymmetric encryption use two keys:

Public Key - to encrypt the data

Private Key - to decrypt the data

- ❖ These keys are generated together.
- ❖ The Public key(s) is distributed freely between the sender and receiver.
- ❖ The other is named as Private Key and it is kept hidden.
- ❖ The Private Key is only used for Decryption and will not be shared between the sender and receiver.

Asymmetric Encryption Algorithms

- ❖ RSA:
- ❖ Diffie-Helman:.
- ❖ Digital Signature Algorithm:

RSA

- by Rivest, Shamir & Adleman of MIT in 1977
- best known & widely used public-key scheme
- based on exponentiation in a finite (Galois) field over integers modulo a prime
 - nb. exponentiation takes $O((\log n)^3)$ operations (easy)
- uses large integers (eg. 1024 bits)
- security due to cost of factoring large numbers
 - nb. factorization takes $O(e^{\log n \log \log n})$ operations (hard)

MERITS & DE-MERITS

Merits:

- ❖ Two parties **don't need to have** their private keys **already shared** in order to communicate using encryption.
- ❖ **Authentication** and **Non-Repudiation** are **possible**. (Authentication means that you can encrypt the message with my public key and only I can decrypt it with my private key. Non-repudiation means that you can "sign" the message with your private key and I can verify that it came from you with your public key.)

De-Merits:

- ❖ Asymmetric Encryption algorithms are comparatively complex.
- ❖ Time consuming process for Encryption and Decryption.

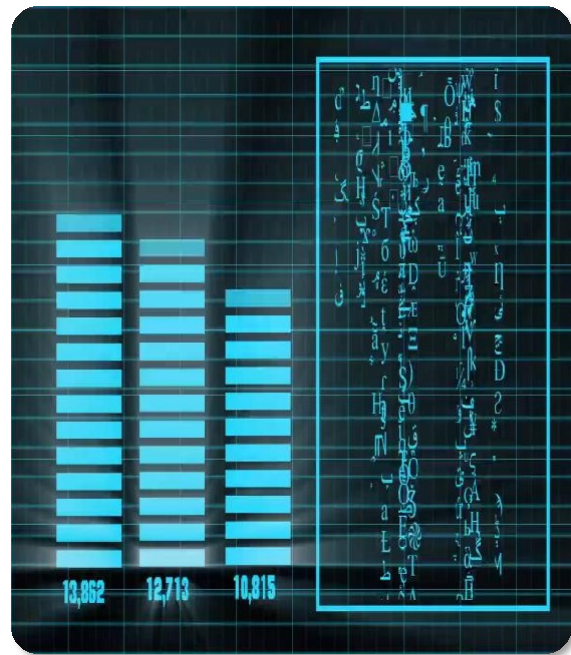
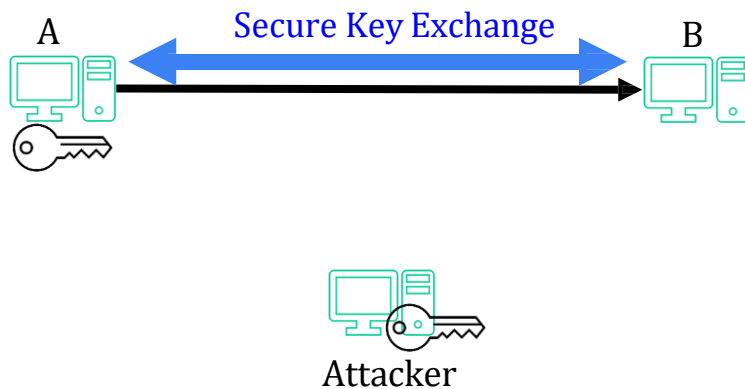
Diffie-Hellman Setup

- all users agree on global parameters:
 - large prime integer or polynomial q
 - a being a primitive root mod q
- each user (eg. A) generates their key
 - chooses a secret key (number): $x_A < q$
 - compute their **public key**: $y_A = a^{x_A} \bmod q$
- each user makes public that key y_A

Digital Signature Standard (DSS)

- US Govt approved signature scheme
- designed by NIST & NSA in early 90's
- published as FIPS-186 in 1991
- revised in 1993, 1996 & then 2000
- uses the SHA hash algorithm
- DSS is the standard, DSA is the algorithm
- FIPS 186-2 (2000) includes alternative RSA & elliptic curve signature variants
- DSA is digital signature only unlike RSA
- is a public-key technique

Diffie-Hellman Key Exchange



Hash functions

Hashing is one-way cryptographic algorithm means that the ciphertext cannot be decrypted to reveal the original plaintext

ABC

Lorem ipsum dolor sit amet,
consectetur adipiscing elit.
Donec at metus at est eleifend
tempus. Phasellus blandit
sollicitudin justo. Placerat,
sem eget consequat sodales.

Plain Text

#

Hash function

#b!c1d
&"(#df
#!sk84#

Hash value

The MD5 hash for hello world is : 5eb63bbbe01eed093cb22b8f5acd3

The MD5 hash for Hello world is : 3e25960a79dbc69b674cd4ec67a72c62



Hash functions (cont.)



Hash functions represent a third cryptography type alongside symmetric and asymmetric cryptography.



A hash function is a function that maps data of any size to data of a fixed size.



The value returned by a hash function is called hash value or simply h which It is often viewed as a digital fingerprint.



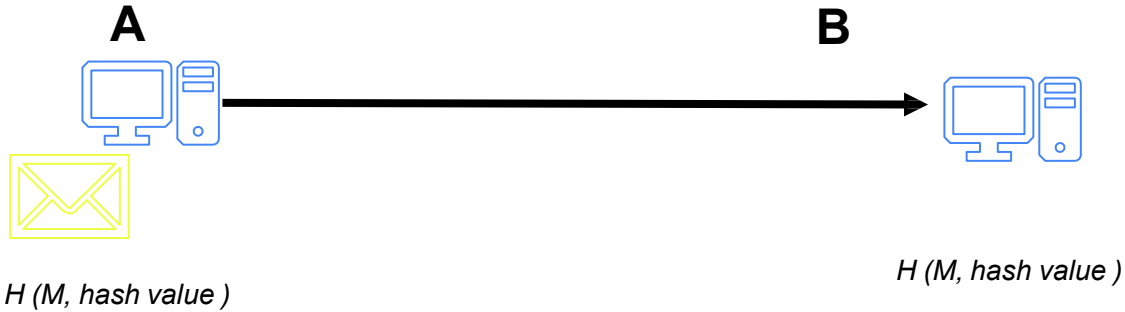
Hashing only provides integrity services; confidentiality services are not supported. A one-way cryptographic algorithm is a hash function. The adoption of a one-way cryptographic algorithm prevents the ciphertext from being decoded to expose the original plaintext.



A hashing function's able to produce completely distinct hash values even when the input plaintext is slightly altered is another noteworthy feature. This characteristic is helpful for data security since it produces clearly different hashes whenever an attacker makes even a small alteration to the original data.



Hash functions (cont.)

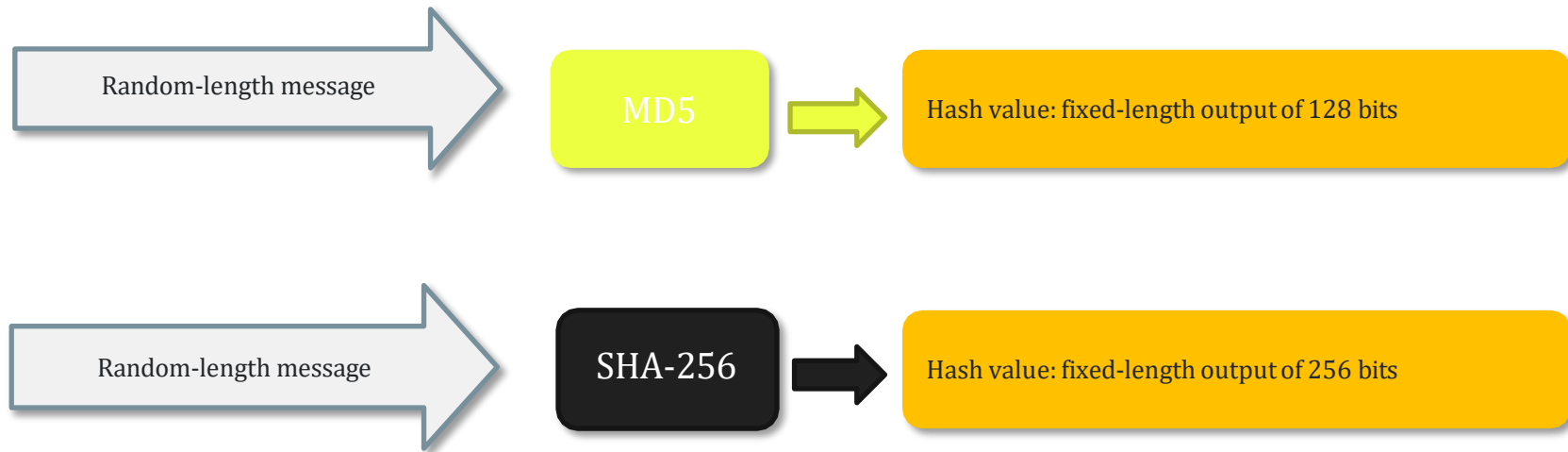


If the hash value generated by A = the hash value generated by B then **integrity** achieved





Common Hash Functions



keyed-hash message authentication code



In cryptography, an HMAC (sometimes expanded as either keyed-hash message authentication code or hash-based message authentication code) is a specific type of message authentication code (MAC) involving a cryptographic hash function and a secret cryptographic key.



As with any MAC, it may be used to simultaneously verify both the data integrity and authenticity of a message.



HMAC can provide authentication using a shared secret instead of using digital signatures with asymmetric cryptography.



It trades off the need for a complex public key infrastructure by delegating the key exchange to the communicating parties, who are responsible for establishing and using a trusted channel to agree on the key prior to communication.

keyed-hash message authentication code (cont.)

The key length plays an important role in hashing algorithm. The bigger the key size, the harder to break in the key while the smaller the key size, the easier to break in the key and the algorithm is more vulnerable to collision attack where the hashing algorithm generates the same hash value for different input



MD5 – Uses a 128-bit shared secret key. The variable-length message and 128-bit shared secret key are combined and run through the HMAC-MD5 hash algorithm. The output is a 128-bit hash. The hash is appended to the original message and forwarded to the remote end.



SHA – SHA-1 uses a 160-bit secret key. The variable-length message and the 160-bit shared secret key are combined and run through the HMAC-SHA1 hash algorithm. The output is a 160-bit hash. The hash is appended to the original message and forwarded to the remote end.

Hashing and encryption differences

Basis	Hashing	Encryption
Definition	It is a process to convert information to a shorter fixed value known as the key that is used to represent the original information.	It is the process to encode data securely such that only the authorized user who knows the key or password is able to retrieve the original data for everyone else it is just garbage.
Purpose	The purpose of hashing is indexing and retrieving items from the database. The process is very fast.	The purpose of encryption is to transform data to keep it secret from others.
Reverse Process	The hash code or key can not be reversed to the original information by any means. It can only be mapped and the hash code is checked if the hash code is the same the information is the same otherwise not. The original information can not be retrieved.	The original information can be easily retrieved if we know the encryption key and algorithm used for encryption.
Secure	It is more secure in comparison to encryption.	It is less secure in comparison to hashing.
Creation of file	Generally, it tries to generate a new key for each information passed to the hash function but on rare occasions, it might generate the same key popularly known as a collision.	It will always generate a new key for each information.
Example	MD5, SHA256	RSA, AES and Blowfish
Output	The output of a hashing algorithm is a fixed-size hash value	the output of an encryption algorithm is ciphertext of the same size or larger than the original data
Length of information	The hashed information is generally of small and fixed length. It does not grow with the increase in the information length of information.	The encrypted information is not of fixed length. It grows with the increase in length of information.
57 key management	Hashing does not require a secret key or algorithm to produce a hash value	encryption requires a secret key or algorithm to encrypt and decrypt data.

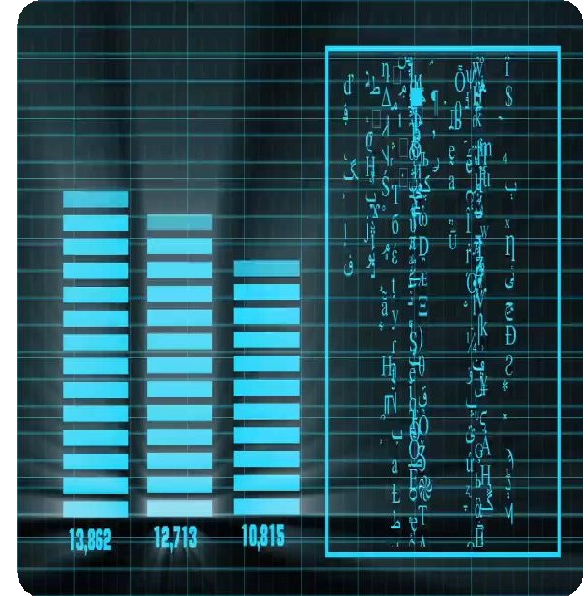


Cryptanalysis attacks

Cryptanalysis= the science of breaking encryption



- To determine the weak points of a cryptographic system, it is important to attack the system. These attacks are called Cryptanalytic attacks.
- The main goal of a Cryptanalysis attack is to recover the key in use rather than simply to recover the plaintext of a single ciphertext.



Cryptanalysis Techniques



Attempt to infer some meaning without even breaking the encryption, such as noticing an unusual frequency of communication or determining something by whether the communication was short or long.



Attempt to deduce the key, in order to break subsequent messages easily

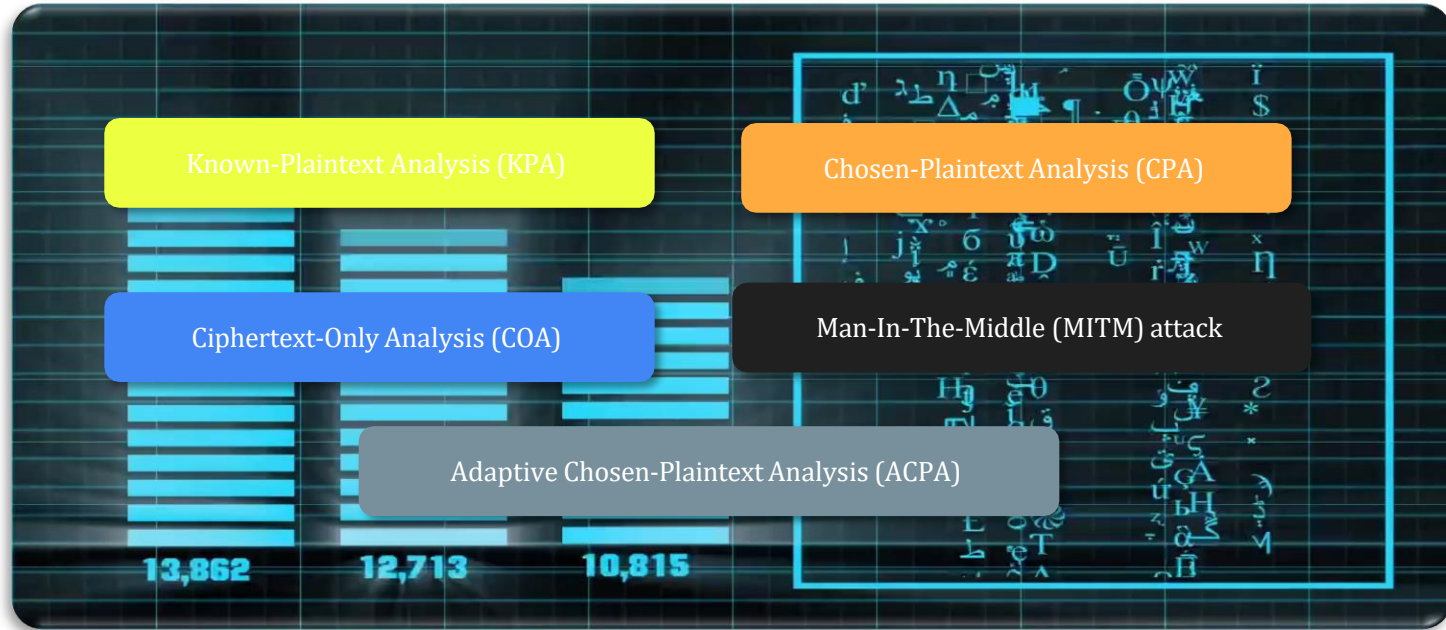


Attempt to find weaknesses in the implementation or environment of the use of encryption.



Attempt to find general weaknesses in an encryption algorithm, without necessarily having intercepted any messages.

Cryptanalytic attacks



Cryptanalytic attacks

- **Known-Plaintext Analysis (KPA) :**
In this type of attack, some plaintext-ciphertext pairs are already known. The attacker maps them in order to find the encryption key. This attack is easier to use as a lot of information is already available.
- **Chosen-Plaintext Analysis (CPA) :**
In this type of attack, the attacker chooses random plaintexts and obtains the corresponding ciphertexts and tries to find the encryption key. It's very simple to implement like KPA but the success rate is quite low.
- **Ciphertext-Only Analysis (COA) :**
In this type of attack, only some ciphertext is known and the attacker tries to find the corresponding encryption key and plaintext. It's the hardest to implement but is the most possible attack as only ciphertext is required.
- **Man-In-The-Middle (MITM) attack :**
In this type of attack, the attacker intercepts the message/key between two communicating parties through a secured channel.
- **Adaptive Chosen-Plaintext Analysis (ACPA) :**
This attack is like CPA. Here, the attacker requests the cipher texts of additional plaintexts after they have ciphertexts for some texts.



References



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Thank you

